SQL Cheat Sheet

1. Basics

```
SELECT column1, column2 FROM table;
SELECT * FROM table;
SELECT DISTINCT column FROM table;
```

2. Filtering

```
SELECT * FROM table WHERE age > 30;
SELECT * FROM table WHERE age > 30 AND city = 'NY';
SELECT * FROM table WHERE country IN ('US', 'UK');
SELECT * FROM table WHERE age BETWEEN 20 AND 30;
SELECT * FROM table WHERE name LIKE 'A%';
```

3. Sorting

```
SELECT * FROM table ORDER BY age DESC;
```

4. Aggregations

```
SELECT COUNT(*) FROM table;
SELECT SUM(salary) FROM employees;
SELECT AVG(price) FROM products;
SELECT MIN(age), MAX(age) FROM users;
SELECT department, AVG(salary) FROM employees GROUP BY department;
SELECT department, AVG(salary) FROM employees GROUP BY department HAVING AVG(salary) > 50000;
```

5. Joins

```
SELECT a.name, b.order_id FROM customers a INNER JOIN orders b ON a.id = b.customer_id;
LEFT JOIN, RIGHT JOIN, FULL JOIN (syntax similar to INNER JOIN)
```

6. Subqueries

```
SELECT name FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
```

7. Set Operations

```
UNION, UNION ALL, INTERSECT, EXCEPT
```

8. DDL (Data Definition)

```
CREATE TABLE employees (id INT PRIMARY KEY, name VARCHAR(100), salary DECIMAL(10,2));
ALTER TABLE employees ADD COLUMN department VARCHAR(50);
DROP TABLE employees;
```

9. DML (Data Manipulation)

```
INSERT INTO employees (id, name, salary) VALUES (1, 'John', 50000);
UPDATE employees SET salary = 60000 WHERE id = 1;
DELETE FROM employees WHERE id = 1;
```

10. Window Functions

```
SELECT name, ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary DESC) AS rn FROM employees;
RANK, DENSE_RANK, LAG, LEAD
```